

Rishi Singhal

PhD in Computer Science, North Carolina State University, Raleigh, USA
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RESEARCH INTERESTS

PhD researcher in **AI safety, interpretability, and alignment**, specializing in analyzing **memorization, privacy leakage, and bias** in large-scale generative models (LLMs, multimodal transformers). My work dissects and interprets different model components at the neuron level via techniques like **quantization, pruning, knowledge distillation**—to explain failure modes and bias, which facilitate in designing scalable cures. I connect **interpretability with efficiency** to ensure robust and trustworthy AI deployment in safety-critical, high-throughput environments. **Available for Summer Internships: May 2026 – Aug 2026.**

EDUCATION

North Carolina State University, Raleigh, NC, USA
PhD in Computer Science, *Advisor: Dr. Jung-Eun Kim*

August 2025 — 2028 (Expected)
Cumulative GPA: 4.00/4.00

North Carolina State University, Raleigh, NC, USA
Master of Science in Computer Science

August 2023 — May 2025
Cumulative GPA: 4.00/4.00

Indraprastha Institute of Information Technology, Delhi, India
Bachelor of Technology in Electronics & Communication

August 2019 — May 2023
Cumulative GPA: 9.00/10.00

SKILLS

- **Programming:** Python, Java, MATLAB, C++, SQL, HTML, CSS, JavaScript
- **Frameworks:** Pytorch, Tensorflow, JAX, Keras, Scikit-Learn, Numpy, Pandas, SpaCy, NLTK
- **Tools:** Flask, Docker, Nvidia Triton, Explainability Tools (GradCam, SHAP, GNNExplainer), Postman, Git, Tableau, PowerBI

RESEARCH EXPERIENCE

Jung Eun Kim's Lab
Graduate Research Assistant

Raleigh, NC, USA
January 2024 — Present

- Discovered novel roles of LayerNorm (LN) in transformers, showing that pruning only **0.1–0.2%** parameters in Post-LN models mitigates **memorization of sensitive/noisy samples by ~70%**, while preserving generalization. On the other hand for Pre-LN models, LN pruning does not mitigate memorization, while degrades generalization - a tradeoff critical for **AI efficiency and generalization**. This observation was verified against both Natural Language Processing and Computer Vision tasks.
- Proved via **gradient-level and theoretical analysis** that **early LN** layers govern memorization & generalization pathways — a key explainability diagnostic tool for understanding intrinsic consistency of model behavior. Paper has been accepted at **NeurIPS 2025**.
- **Ongoing:** Studying impact of quantization and LoRA on memorization, separating memorization vs. generalization at the feature level, analyzing how residual pathways influence privacy and memorization in large-scale models (GPT, LLaMA).

MIDAS Lab
Undergraduate Research Assistant

New Delhi, Delhi, India
January 2022 — April 2023

- Explored **document coherence** as a fundamental quality metric in NLP tasks (Summarization, Machine Translation, Question-Answering), improving accessibility and naturalness of AI-generated text.
- Applied **Topological Data Analysis** on transformer attention graphs, capturing structural patterns beyond token-level modeling.
- Built lightweight single-layer models incorporating TDA features, that outperformed transformer baselines by **5% accuracy scores** on well-known GCDC and Wikipedia datasets. Paper has been accepted at **ICLR 2024**.

WORK EXPERIENCE

Fermi National Accelerator Laboratory (Fermilab)
Machine Learning Engineer

Batavia, Illinois, USA
May 2024 — August 2024

- Deployed Graph Neural Networks (NuGraph2 and NuGraph3) for real-time background filtering and semantic labeling of primary physics interactions within the MicroBoone project.
- Utilized Nvidia's Triton Inference Server for GPU-as-a-Service deployment on Fermilab's EAF server docker container, leveraging a Python-backend setup to handle raw data ingestion, preprocessing into graph structures, inference, and output delivery.
- Developed client-side code in Python and C++ for diverse use cases, integrating the LarSoft framework to streamline data transmission directly from the client to the server, by reducing memory overhead by **20%** while bypassing the need for intermediate H5 files.
- Implemented the NuSonic Triton framework to simplify user interaction with Triton services and enhance code maintenance by providing a unified interface and reducing dependency on API changes.

- Thoroughly documented code to ensure reproducibility and ease of further development, supporting ongoing work for the Micro-Boone project and future advancements in the DUNE experiment.

MITACS Globalink Research Internship(Concordia University)
Research Intern

Montreal, Quebec, Canada
 May 2021 — August 2021

- Worked as a research intern with Dr. Jun Yan in the CIISE department at Concordia University, contributing to the analysis and synthesis of research in the field of cybersecurity and Machine Learning.
- Surveyed and analyzed research papers on intrusion detection systems and smart grids, concentrating on data analytics and modeling.
- Collaborated with Ericsson and MITACS to define essential requirements for these systems, including performance, generalizability, feasibility, robustness, and explainability.

PUBLICATIONS

Published Papers

- **Rishi Singhal**, Krishna, S., Singla, Y.K. and Shah, R.R., Impact of Layer Norm on Memorization and Generalization in Transformers. **Accepted At: Advances in Neural Information Processing Systems (NeurIPS), 2025 - A* Conference (Top 25%)**.
- **Rishi Singhal**, Samyak Jain, Sriram Krishna, Yaman Kumar Singla, Rajiv Ratn Shah, Beyond Words: A Topological Exploration of Coherence in Text Documents. **Accepted At: The Second Tiny Papers Track at ICLR 2024 - A* Conference**.
- **Rishi Singhal**, Shailender Gupta, Poonam Singhal. An Ensembling Approach using DL & Non-DL Techniques for Detecting Brain Tumors using MRI Scans. **Accepted At: 2022 IEEE International Power and Renewable Energy Conference (IPRECON)**.
- **Rishi Singhal**, Poonam Singhal, Shailender Gupta, Solar-Cast: Solar Power Generation Prediction from Weather Forecasts using Machine Learning. **Accepted At: 2022 IEEE 10th Power India International Conference (PIICON)**.

CONFERENCE TALKS AND POSTERS

- **An Ensembling Approach using DL & Non-DL Techniques for Detecting Brain Tumors using MRI Scans**
 Conference Talk, PIICON 2022 - New Delhi, India, 2022
- **Solar-Cast: Solar Power Generation Prediction from Weather Forecasts using Machine Learning**
 Conference Talk, IPRECON 2022 - Kerala, India, 2022

AWARDS

NeurIPS 2025 Scholar Award San Diego, CA, USA
 Honored as a NeurIPS 2025 Scholar for contributions to AI research; included a fully funded conference attendance. October 2025

NC State University COE Academic Travel Grant Raleigh, NC, USA
 Awarded an academic travel grant by the College of Engineering to support conference participation. October 2025

Data Science (Data Unchained) Hackathon New Delhi, India
 Secured 2nd position in the Data Science Hackathon organized at IIIT Delhi by developing a robust LSTM-RNN model for time series medical data. January 2022

Game Development Hackathon New Delhi, India
 Finished 3rd while participating in the Game Development Hackathon as part of Technical Fest Esya'21 at IIIT Delhi. September 2021.

PROJECTS

Exploring & Analyzing Internal Structure of Language Models to Mitigate Social Biases Raleigh, NC
Link, Techstack Used: Python, Pytorch August 2024 — December 2024
 Investigated the encoding of social biases (e.g., gender, race, sexual orientation) within Pre-trained Language Models (PLMs) like BERT, RoBERTa. Analyzed how, when, and where these biases are embedded in the model's internal layers and neurons, revealing **their presence in later model layers and the Feed-Forward Network instead of Attention Heads**. This research aims to inform better mitigation strategies like pruning without relying solely on costly dataset curation or fine-tuning.

Few Informative Data Samples are Good Enough: Introducing Intelligent Data Pruning Raleigh, NC
Link, Techstack Used: Python, Scikit-Learn, Numpy, Pandas February 2024 — April 2024
 Developed an 'intelligent data pruning' methodology for under-sampling class-imbalanced datasets, improving model efficiency and accuracy. Outperformed traditional and advanced sampling methods like SMOTE, Gaussian Copula, SDV, RRP, across various software engineering datasets and models, while offering practical insights for large-scale datasets with limited resources.

Out-of-Distribution Detection using Uncertainty Predictions in Quantized Networks Raleigh, NC
Link, Techstack Used: Python, Tensorflow, Numpy August 2023 — November 2023

Innovated a technique using Monte-Carlo Dropouts for Out-of-Distribution Detection, i.e., uncertainty quantification in Quantized Networks, achieving superior accuracy and F1-Score on CIFAR100 and CIFAR100 C datasets compared to existing models like ResNet50, VGG-16. Additionally, introduced a novel metric through a comprehensive visual analysis for enhanced result interpretation.

PopcornPicks: Movie Recommendation System

Raleigh, NC

Link, Techstack Used: Python, Flask, Collaborative Filtering, HTML, CSS, JavaScript

August 2023 — November 2023

Engineered PopcornPicks, a movie recommender system utilizing Python, Flask, Collaborative Filtering, HTML, CSS, and JavaScript. Delivered efficient and adaptable movie suggestions based on user-input preferences, showcasing expertise in collaborative filtering algorithms and full-stack development.

Generalized Cross Multi-Domains Representations using User-Sentiments

New Delhi, Delhi

Link, Techstack Used: Python, Pytorch, Numpy

February 2023 — April 2023

Implemented a recommendation system for generating generalized multi-cross-domain representations using user sentiments, addressing limitations in two-domain-focused approaches. Utilized item adjacency list (T) and user metadata, significantly improving cross-domain recommendation system performance with enhanced NDCG and HIT metrics.

TEACHING EXPERIENCE

Graduate Teaching Assistant

August 2025

- Teaching Assistant in CSC510 Software Engineering course at NC State University, Raleigh, under Dr. Timothy Menizes.

Data Science Mentor at ASA Datafest, NCSU

April 2024

- Guided and mentored aspiring data scientists in analyzing real-world datasets during the 48-hour hackathon.
- Provided expertise in data science and analytics, fostering creativity and problem-solving among participants.

Undergraduate Teaching Assistant

January 2023 — April 2023

- Managed sessions, assignments, and quizzes and tutored students in the Computer Vision course at IIIT Delhi.
- Taught about developing deep-learning models like VGGNet, ResNet, AlexNet and FCN-ResNet50 and how to finetune them for computer vision tasks of classification, object detection and segmentation.

Data Science-Computational Biology Club(BioBytes) Coordinator

January 2021 — April 2022

- Led the Data Science-Computational Biology club at IIIT Delhi where by holding and managing various data science and machine learning sessions intersection with computational biology.

Student Mentorship Program at IIIT Delhi

April 2022 — April 2023

- Guided and helped group of 4 mentees in freshman year about the academic, work and personal culture at IIIT Delhi.